

Nihal Veeramalla

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SUMMARY

Data Scientist specializing in large language models, RAG systems, and production AI infrastructure. Expert in LLM fine-tuning, vector databases, and building scalable AI applications with proven experience deploying transformer models at enterprise scale. Strong background in MLOps, distributed training, and AI system optimization with focus on performance, security, and reliability in production environments.

EXPERIENCE

Fiserv

Jan 2024 - Present

Data Scientist / AI

- Built a tenant-isolated AI agent and RAG platform on Azure OpenAI and Azure AI Search with hybrid retrieval using BM25, dense vectors, and cross-encoder reranking, delivering natural-language analytics on financial policies and transactions and reducing analyst research time by 80%.
- Architected multi-agent systems with LangChain and LangGraph using tool and function calling for Microsoft Fabric SQL, KQL, SharePoint search, Microsoft Graph, and ticketing; added JSON Schema input and output, self-verification and re-ask loops, and deterministic parsing, achieving 92% valid SQL and KQL on complex analytical questions.
- Implemented end-to-end ingestion pipelines with Azure Data Factory and Azure Functions including scheduled crawlers, layout-aware parsing for PDFs, HTML, and CSV, text normalization, PII detection and redaction, document versioning, lineage capture, and rich metadata for tenant and access control with storage in Microsoft Fabric OneLake Delta.
- Designed advanced chunking strategies including heading-aware recursive splitting with windowed overlap, semantic windowing for long context, table and figure handling, and multi-vector embeddings for titles, body text, and tables; improved grounding coverage and answer faithfulness across large policy and transaction corpora.
- Indexed Fabric tables via SQL and KQL endpoints and SharePoint and OneDrive content into Azure AI Search vector indexes; enabled unified hybrid lexical and dense retrieval, late fusion across policy and transaction stores, multi-hop reasoning, and kept P95 retrieval latency under 100 milliseconds at over one million embeddings.
- Orchestrated router retrievers that classify user intent and route to the optimal index and strategy; implemented query rewriting and expansion, per-tenant lexical and dense fallbacks, and guardrails for safe and auditable tool selection.
- Standardized agent tool integrations using Model Context Protocol and FastMCP servers; created a permissioned tool registry with role-based access control, managed identities, and scoped secrets, exposing Fabric SQL, KQL, SharePoint, Microsoft Graph, and ticketing tools to LangChain and LangGraph agents and reducing onboarding time for new tools.
- Integrated deeply with the Microsoft ecosystem by enforcing Entra ID permissions and Graph scopes, aligning access models across Fabric, SharePoint, and agent tools, and exposing webhook endpoints for Power Platform automation and approvals.
- Established MLOps with Docker containers, Azure Kubernetes Service, infrastructure as code using Bicep and Terraform, and continuous integration and continuous delivery with GitHub Actions and Azure DevOps; delivered blue-green and canary releases, autoscaling, health probes, and supported more than 100 QPS with P95 routing latency under 200 milliseconds and 99.8% uptime.
- Built observability and evaluation workflows with LangSmith for tracing and telemetry and with Ragas and DeepEval for quality and robustness; added automated red-team suites and promotion gates and consolidated runtime metrics in Azure Monitor and Application Insights, reducing hallucinations by 35% before release.
- Enforced security, compliance, and privacy with prompt-injection and data-exfiltration filters, SQL and KQL guardrails, role-based access control with Entra ID and Privileged Identity Management, secret management in Azure Key Vault, private endpoints and virtual network isolation, and immutable audit logging aligned to PCI and SOC 2.
- Optimized latency and cost using FastAPI with asynchronous processing, Azure Cache for Redis for semantic and result caching, server-sent events streaming, prompt templating and truncation, selective function calling, and intelligent batching; reduced P95 latency from 4.2 seconds to 1.3 seconds and token spend by 30%.
- Defined agent behavior specifications with success criteria, invariants, safe states, allowed tool scopes, and a mapped catalog of edge cases including ambiguity, low context, conflicting instructions, and stale data.
- Developed an agent test harness using LangSmith traces and scenario matrices and implemented golden datasets and property-based tests for tool-call idempotency, loop escape, and deterministic parsing.
- Ran offline workflow evaluations with Ragas measuring faithfulness, answer relevance, context recall, and answer similarity and tracked task success rate, step-level latency budgets, and tool success codes across critical scenarios.
- Implemented automated red-team testing for prompt injection, data exfiltration, tool abuse, role confusion, and jailbreak attempts and enforced deployment gates based on evaluation thresholds and safety criteria.
- Added agent guardrails including tool-usage policies, quotas and timeouts, retry with backoff, circuit breakers, confidence scoring, and human-in-the-loop escalation on low confidence or safety triggers.
- Enabled replay and regression testing from production traces in LangSmith to validate fixes, prevent drift, and verify improvements prior to promotion to higher-traffic tiers.
- Delivered hybrid retrieval on Azure AI Search vector indexes in production and evaluated PostgreSQL with pgvector for prototyping and data-co-located embeddings to benchmark cost and performance trade-offs.
- Authored model cards, behavior reports, runbooks, and experiment summaries that document evaluation methodology, safety decisions, and known limitations and drove cross-functional knowledge sharing with engineering, product, and design.

Hyperleap AI

Jun 2020 - Aug 2022

Data Scientist

- Built a customer segmentation platform on PySpark (AWS EMR/Glue) using behavioral features and clustering (K-Means/DBSCAN), improved targeting accuracy 22% and increased Silhouette Score by 0.14, enabling segment-specific campaigns and lift analysis.

- Designed a multi-stage data foundation: real-time and batch ETL (Glue/EMR/Lambda), schema-drift detection, data contracts, quality checks (completeness, uniqueness, freshness), and a curated Redshift warehouse powering analytics and ML; reduced pipeline latency 63% and improved data reliability 25%.
- Implemented closed-loop personalization: segment rollouts with A/B testing and uplift modeling; used Z-tests, Mann-Whitney U, and retention (Kaplan-Meier) to prove an 18% engagement uplift and operationalize winner selection.
- Developed a catalog vision + OCR pipeline for image cleanup, product detection and normalization, barcode/label parsing, and entity extraction into a governed schema, reduced manual catalog edits 70% and achieved ~85% field-level accuracy.
- Built anomaly detection on SageMaker to filter 12% outlier customers via DBSCAN hyper-parameter optimization, improving downstream model stability and conversion.
- Stood up MLOps: model registry and feature store patterns, CI/CD with blue-green deployment, drift monitoring (data/feature/prediction), and automated rollback triggers accelerated release cycles 50% and cut incident MTTR 60%.
- Delivered governance & security for analytics/ML role-based access policies, PII tokenization, audit trails, and cost observability; lowered monthly compute/storage cost 28% through right-sizing and spot-aware scheduling.
- Built executive analytics (Power BI + Redshift) with topic modeling and sentiment to surface segment insights, campaign saturation, and incremental ROI, reduced manual analysis time 70% across marketing and product teams.

Infinite Infolab

Jan 2019 - Jun 2020

Data Scientist

- Worked on a hierarchical document classification system with a 3-tier ML architecture processing 30+ document types, implementing intelligent preprocessing and feature extraction to reduce manual review by 40%.
- Built anomaly detection layer using Isolation Forest on SageMaker to identify out-of-distribution documents, filtering ~8% anomalous inputs and improving downstream classifier precision by 15%.
- Developed production document classifier using Random Forest with optimized TF-IDF features on EMR Spark, processing 10M+ documents across multiple categories with 95% precision and sub-50ms inference latency.
- Implemented advanced hierarchical attention networks using BERT embeddings and TensorFlow, deployed via SageMaker real-time endpoints and improving multi-class F1-score by 18% across 50+ document subtypes.
- Built a scalable data lake on S3 with intelligent partitioning and lifecycle management, reducing I/O latency by 30% and enabling seamless integration with ML training and inference pipelines.
- Optimized model experimentation using EC2 GPU instances with auto-scaling groups, accelerating PyTorch prototyping by 40% while reducing compute costs by 25% through strategic spot instance usage.
- Automated ML workflows using serverless Lambda functions for model retraining triggers, reducing manual intervention by 80% and ensuring continuous model improvement with real-time data ingestion.
- Implemented high-throughput batch inference using SageMaker Batch Transform for processing 100K+ documents nightly, improving operational efficiency by 50% and reducing processing costs by 60%.
- Established robust CI/CD pipelines using MLflow, Docker, and Git-based workflows for reproducible model deployment, reducing release cycles by 30% and enabling rapid rollback capabilities.
- Built comprehensive production monitoring with CloudWatch integration and custom performance metrics, implementing automated alerting for model degradation with sub-24h drift detection and remediation.

EDUCATION

George Mason University

Aug 2022 - May 2024

Master of Science, Data Analytics Engineering

IIIT-Bangalore

Nov 2020 - Dec 2021

Postgraduate, Data Science with Specialization in Deep Learning

SRM University

Jun 2016 - May 2020

Bachelor of Technology, Computer Science and Engineering

Publications

- International Journal of Advanced Science and Technology .Detection and Prevention of DDoS Attack in SDN.

SKILLS

- **Programming Languages:** Python, R, Java, SQL
- **Machine Learning Frameworks & Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, TensorFlow, Keras, PyTorch, NLTK, OpenCV, XGBoost, LightGBM, CatBoost, Beautiful Soup
- **Artificial Intelligence & NLP:** Generative AI, Large Language Models (LLMs) BERT, GPT-3/4, RoBERTa, T5, Transformers, Named Entity Recognition (NER), Topic Modeling (LDA), Prompt Engineering, RAG (Retrieval Augmented Generation)
- **Machine Learning Algorithms:** Linear Regression, Decision Trees, SVM, Random Forests, KNN, K-Means, CNN, LSTM, Logistic Regression, DBSCAN, Naive Bayes, A/B Testing, Ensemble Methods
- **Deep Learning Architectures:** CNNs, RNNs, LSTMs, GANs, Transformers, Hierarchical Attention Networks (HAN)
- **Data Processing & ETL:** Apache Spark, Apache Kafka, Airflow, zure Data Factory, AWS Glue
- **Big Data Technologies:** Hadoop, HDFS, Hive, Spark, Snowflake, Presto
- **Cloud Platforms & Services:** AWS (S3, EC2, Lambda, SageMaker, Bedrock, CloudWatch), Google Cloud Platform (BigQuery, Dataflow, Vertex AI), Microsoft Azure (Synapse Analytics, Data Lake, Azure ML)
- **Data Visualization Tools:** Tableau, Power BI, Looker, Plotly, Matplotlib, Seaborn
- **Databases:** MySQL, PostgreSQL, SQL Server, Oracle, MongoDB, Cassandra, DynamoDB, Redis
- **Model Deployment & MLOps:** Flask, FastAPI, Docker, Kubernetes, MLflow, Terraform, Jenkins, CI/CD Pipelines
- **Version Control & Collaboration:** Git, GitHub, GitLab, Bitbucket, Jupyter Notebooks, Google Colab, VS Code